

Jannes Hühnerbein

PhD Candidate @ Technical University of Munich | Robotics, Systems and Control MSc ETH

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Education

Technical University of Munich — Chair of Astrodynamics

Ottobrunn/Taufkirchen, Germany

DOCTORATE | AEROSPACE ENGINEERING

since January 2026

- Research focused on dynamical systems and control problems for space applications

ETH Zurich / Swiss Federal Institute of Technology Zurich

Zurich, Switzerland

ROBOTICS, SYSTEMS AND CONTROL MSc ETH

Sep 2022 — Nov 2024

- *Master's Thesis*: Uncertainty Model Unfalsification via Semi-Infinite Programming and Local Reduction
- *Semester Project*: Diffusion Spline-Based Navigation Policy in Dynamic Environments
- *Core Courses*: Linear System Theory, Dynamic Programming and Optimal Control, Recursive Estimation, Model Predictive Control, Vision Algorithms for Mobile Robotics, Nonlinear Dynamics and Chaos, Introduction to Machine Learning
- *Additional Courses*: Advanced Model Predictive Control, Convex Optimization, Probabilistic Artificial Intelligence, Orbital Dynamics

MECHANICAL ENGINEERING BSc ETH

Sep 2018 — Oct 2021

- *Thesis*: Distributed Trajectory Planning for Multiple Autonomous Aerial Vehicles
- *Core Courses*: Calculus, Linear Algebra, Mechanics, Dynamics, Physics, Thermodynamics, Fluid Mechanics, Electrical Engineering, Informatics, Control Systems
- *Electives*: Signals & Systems, System Modeling, Models, Algorithms and Data

Gymnasium Raabeschule

Braunschweig, Germany

ABITUR (GERMAN UNIVERSITY-ENTRANCE QUALIFICATION)

graduated in June 2017

- Salutatorian
- *Advanced Courses* (German “Leistungskurse”): Mathematics, Physics, Chemistry

Research & Work Experience

Chair of Astrodynamics, Technical University of Munich

Munich, Germany

SCIENTIFIC ASSISTANT (WISSENSCHAFTLICHER MITARBEITER)

since July 2025

- Contributing to teaching activities, including preparing course materials, supervising student exercises and conducting examinations
- Supervising Bachelor's, Master's and semester theses for students of the Aerospace Engineering degree programs
- Supporting research activities and laboratory development
- 📄 Chair website

Control and Power Research Group, Imperial College London

London, United Kingdom

MASTER'S THESIS

Apr 2024 — Nov 2024

- *Title*: Uncertainty Model Unfalsification via Semi-Infinite Programming and Local Reduction
- Conducted as a visiting student under supervision of Prof. Dr Eric Kerrigan
- Explored the use of optimization problems (semi-infinite programs in particular) and data to inform uncertainty models
- Developed a Julia package for dynamical systems simulation called 📄 SimpleSim.jl available on the official Julia registry

Robotic Systems Lab, ETH Zurich

Zurich, Switzerland

SEMESTER PROJECT

Oct 2023 — Feb 2024

- *Title*: Diffusion Spline-based Navigation Policy in Dynamic Environments
- Investigated the use of denoising diffusion probabilistic models (DDPMs) for spline-based robot navigation

Zipline International Inc.

South San Francisco, California, USA

GUIDANCE, NAVIGATION AND CONTROL INTERN

Sep 2021 — Aug 2022

- Supported development work on guidance, navigation and controls algorithms for a novel autonomous aircraft
- Worked extensively with Julia (for simulation) and C++ (for prototyping and embedded systems)
- 📄 What I worked on (YouTube)

Institute for Dynamic Systems and Control, ETH Zurich

Zurich, Switzerland

BACHELOR'S THESIS

Summer 2021

- *Title*: Distributed Trajectory Planning for Multiple Autonomous Aerial Vehicles
- Developed a distributed trajectory planner using ADMM and ALADIN and contributed to the 📄 CRS framework

Academic Space Initiative Switzerland (ARIS)

Zurich & Dübendorf, Switzerland

CONTROL SYSTEMS ENGINEER

Sep 2020 — Aug 2021

- Two-semester project with the goal of developing an autonomous guided recovery system for a student rocket
- Worked with a team of eight students from different fields and universities
- Developed a fully functional embedded control system from the ground up
- 📄 Link to ARIS Website

- Independently taught weekly classes of approx. 25 students
- Prepared practice material and exercise sessions
- Supported lectures by preparing slides, grading assignments and monitoring online forums

Honors & Awards

2020 — 2024	Scholarship Recipient  More Information in English	German National Academic Foundation
May 2019	1st Place Innovation Project (“Innovationsprojekt”),  Information in German	ETH Zurich
Jul 2017	High-School Graduate Prize  More Information in German	German Physical Society

Specialized Skills

Programming	Julia, Python, C++, C
Robotics	MATLAB/Simulink, ROS, Gazebo, Eigen
Data Science & ML	NumPy, PyTorch, Pandas
Optimization	JuMP, CasADi, FORCESPRO, Acados, CVXPY
Development	Linux, Git, Docker, Network Administration
Prototyping	CAD, 3D Printing, RaspberryPi/Arduino
Embedded Hardware	STM32, ESP32, Sensor Integration, I2C, SPI, UART, KiCAD
Other	LaTeX, SQL, Web Development, IT Administration (...)

Other

Languages	German — Native Proficiency (C2) English — Full Professional Proficiency (C2) French — Elementary Proficiency (A2)
Volunteering	Mentoring-Program for First-Semester Students at ETH Zurich 2020/2021 Advisor for Digital Infrastructure & IT at Youth Against AIDS (Jugend gegen AIDS e. V.) 2014 — 2017